Relationship Satisfaction Analysis in Australia
From a survey provided by the Centre for Australian Relationship Excellence (CARE)

GROUP62_01

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Executive Summary

- Comprehensive Analysis: This report analyses relationship satisfaction data collected in 2024 from 700 Australian residents to examine how demographic factors influence relationship satisfaction.

- Significant Findings: The findings show that there is insignificant evidence to suggest the influence of relationship duration, migrancy status and urban or rural residence on relationship satisfaction.

- Recommendations: The Centre for Australian Relationship Excellence should focus on gender, age, conflict, time together per week, and marriage status to critically analyse key factors that affect relationship satisfaction.

Introduction

This report analyses factors influencing relationship satisfaction in Australia, following the 2024 national survey conducted by the Centre for Australian Relationship Excellence (CARE). We will provide a comprehensive investigation and analysis of relationship satisfaction levels, focusing on major potential influences, such as gender, urban or rural living location, age, years in relationship, number of conflicts, time spent together, presence of children, marital status, and migrancy background. The report includes a descriptive analysis of satisfaction, an estimated confidence interval of the difference in satisfaction facilitated by relationship duration, a hypothesis test on the satisfaction of migrants, and a simple regression analysis to evaluate the effects of urbanisation on relationships.

The findings suggest that while most individuals report moderately high satisfaction, migrant backgrounds and urban-rural locations do not have a significant impact on relationship satisfaction among Australians. The utilisation of multiple regression analysis demonstrates that gender, age, conflicts, time spent together per week, and marriage factors appear to have strong associations with the variation of satisfaction levels. All figures in this report are rounded to 4 decimal places.

Analysis

A. Descriptive Analysis

- Gender: Male=1, Female=0

- Children: Yes=1, No=0

- Married: Yes=1, No=0

- Migrant: Yes=1, No=0

- Urban=1, Rural=0

Satisfaction	
Mean	7.0042
Standard Error	0.0474
Median	7.02
Mode	6.51
Standard Deviation	1.2534
Sample Variance	1.5710
Kurtosis	-0.0556
Skewness	-0.1482
Range	7.33
Minimum	2.67
Maximum	10
Sum	4902.97
Count	700

Figure 1: Satisfaction

Gender	
Mean	0.4771
Standard Error	0.0189
Median	0
Mode	0
Standard Deviation	0.4998
Sample Variance	0.2498
Kurtosis	-1.9973
Skewness	0.0917
Range	1
Minimum	0
Maximum	1
Sum	334
Count	700

Figure 2: Gender

Urban	
Mean	0.5214
Standard Error	0.0189
Median	1
Mode	1
Standard Deviation	0.4999
Sample Variance	0.2499
Kurtosis	-1.9983
Skewness	-0.0860
Range	1
Minimum	0
Maximum	1
Sum	365
Count	700

Figure 3: Urban

Age	
Mean	29.8943
Standard Error	0.1886
Median	30
Mode	31
Standard Deviation	4.9899
Sample Variance	24.8987
Kurtosis	-0.0441
Skewness	-0.0108
Range	28
Minimum	15
Maximum	43
Sum	20926
Count	700

Figure 4: Age

Years in Relationship	
Mean	4.1173
Standard Error	0.0722
Median	4
Mode	3.6
Standard Deviation	1.9091
Sample Variance	3.6448
Kurtosis	-0.3720
Skewness	0.1720
Range	10.3
Minimum	0.1
Maximum	10.4
Sum	2882.1
Count	700

Figure 5: Years in Relationship

Conflict	
Mean	2.0129
Standard Error	0.0534
Median	2
Mode	2
Standard Deviation	1.4116
Sample Variance	1.9927
Kurtosis	0.1131
Skewness	0.6076
Range	7
Minimum	0
Maximum	7
Sum	1409
Count	700

Figure 6: Conflicts in a week

Time together per week	
Mean	14.8561
Standard Error	0.1847
Median	14.5
Mode	14.5
Standard Deviation	4.8871
Sample Variance	23.8838
Kurtosis	-0.0930
Skewness	0.0863
Range	27.7
Minimum	0.5
Maximum	28.2
Sum	10399.3
Count	700

Figure 7: Hours together per week

Children	
Mean	0.3129
Standard Error	0.0175
Median	0
Mode	0
Standard Deviation	0.4640
Sample Variance	0.2153
Kurtosis	-1.3494
Skewness	0.8090
Range	1
Minimum	0
Maximum	1
Sum	219
Count	700

Figure 8: Children

Married	
Mean	0.4029
Standard Error	0.0186
Median	0
Mode	0
Standard Deviation	0.4908
Sample Variance	0.2409
Kurtosis	-1.8477
Skewness	0.3970
Range	1
Minimum	0
Maximum	1
Sum	282
Count	700

Figure 9: Marriage

Migrant	
Mean	0.2557
Standard Error	0.0165
Median	0
Mode	0
Standard Deviation	0.4366
Sample Variance	0.1906
Kurtosis	-0.7425
Skewness	1.1223
Range	1
Minimum	0
Maximum	1
Sum	179
Count	700

Figure 10: Migrancy status

<u>Central Tendency:</u>

As gender, urban/rural, children, marriage and migrancy status are binary variables, we focus on the remaining variables, which provides a more meaningful insight into the influences associated with relationship satisfaction.

- **Age**: a nearly symmetric distribution with a mean of 29.8943, median of 30 and mode of 21, indicates that most participants are in their late 20s to early 30s. The standard deviation is 4.9899, demonstrating a moderate spread. This is further evidenced by a 28 year age range, with a sample variance of 24.8987.
- **Years in relationship**: a fairly symmetric distribution with a mean of 4.1173, median of 4 years and mode of 3.6, indicates a short duration. The standard deviation is 1.9091, a moderate spread of data. The 10.3 year gap indicates a wide variability in relationship duration.
- **Conflicts**: a symmetric distribution with average conflict frequency is 2.0129, median and mode at 2. The standard deviation of 1.4116, a moderate spread.

- **Time together per week:** a symmetrical distribution, with an average time spent 14.8561 hours, median and mode at 14.5 hours per week. The standard deviation of 4.8871 hours, a moderate spread. The skewness is 0.086, close to zero, almost no skew.

Focused Analysis: Satisfaction

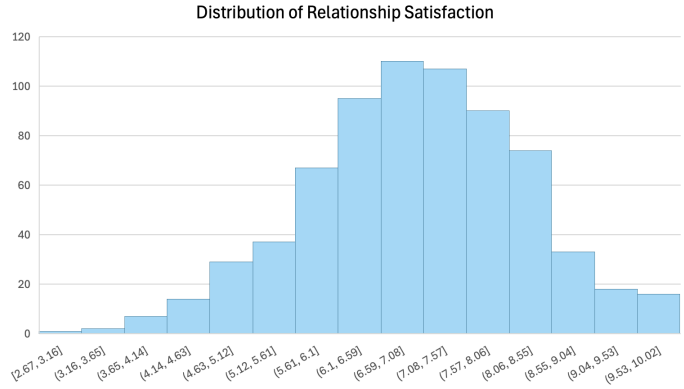

Figure 11: Distribution of Relationship Satisfaction

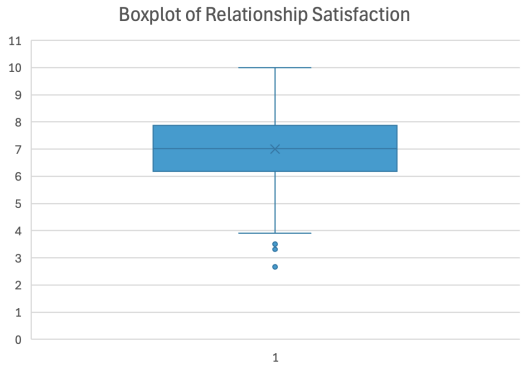

Figure 12: Boxplot

The mean satisfaction score in this dataset is 7.0042, suggesting that, on average, individuals report moderately high satisfaction relationships. The median (7.02) is slightly higher than the mean, indicating that at least half of the participants rate their satisfaction levels at just above the average. The mode (6.51) demonstrates that most individuals' satisfaction levels are slightly below the average, contributing to a slight left skew in the distribution.

Spread and Variability:

The standard deviation (1.2534) indicates a moderate degree of variation in relationship satisfaction scores among participants. The range (7.33) shows that there is a relatively wide spread in how individuals perceive their relationship satisfaction. While the average is high, the range highlights that there is a minority of respondents who report low levels of satisfaction.

Shape of the Distribution:

A slightly negative skew of -0.1482 shows that the distribution is slightly left-skewed, meaning most satisfaction rates are at the higher end, with a few lower scores extending the left tail. As kurtosis (0.0556) is close to zero, the distribution is platykurtic, suggesting fewer extreme values or outliers in the data.

Key observations: The distribution of relationship satisfaction scores suggests consistency, with most individuals reporting moderately high satisfaction. The small difference between the mean and median, along with a mode slightly lower than the mean and median, indicates a slight negative skew. This implies that while on average, individuals experience high levels of relationship satisfaction, there is a minority reporting lower scores. The negative skew and near-zero kurtosis further indicate that the distribution is close to normal, with minimal extreme values or outliers.

B. The 7-Year Itch Hypothesis

The 7-Year Itch Hypothesis theories that relationship satisfaction declines after approximately 7 years. To evaluate the validity of the hypothesis, we divided the sample into two groups: Group 1: individuals in relationships for 7 years or fewer, and Group 2: individuals in relationships for more than 7 years. Based on the sample given of 700 Australians, the Central Limit Theorem of normal distribution can be implemented. We use F-test to assume equal or unequal variances.

F-Test Two-Sample for Variances		
	7 years of fewer	*More than 7 years*
Mean	3.817465224	7.777358491
Variance	2.717852388	0.458323657
Observations	647	53
df	646	52
F	5.929984944	
P(F<=f) one-tail	3.53636E-12	
F Critical one-tail	1.4418423	

Figure 13: F-Test Two-Sample for Variances

According to the F-test, the F Critical one-tail is 1.4418, which is smaller than the F value of 5.930. In addition, the P-value is significantly small, and can be considered as 0.00, which is smaller than 0.05 at 95% confidence. Therefore, there is sufficient evidence that the two variances are unequal. As the population standard deviation is unknown, we conducted a two-sample t-test, assuming unequal variances.

t-Test: Two-Sample Assuming Unequal Variances		
	7 years of fewer	*More than 7 years*
Mean	3.817465224	7.777358491
Variance	2.717852388	0.458323657
Observations	647	53
Hypothesized Mean Difference	0	
df	113	
t Stat	-34.93497091	
P(T<=t) one-tail	1.07151E-62	
t Critical one-tail	1.658450216	
P(T<=t) two-tail	2.14302E-62	
t Critical two-tail	1.981180359	

Figure 14: t-Test: Two-Sample Assuming Unequal Variances

When performing T-test, the null hypothesis is H_0: there is no difference in satisfaction between two groups. It can be observed that the P-Value is significantly lower than 0.05 at 95% confidence. As a result, we use the confidence interval formula that assumes population variances are unknown and unknown equal as follow:

$$CI = \overline{X_1} - \overline{X_2} \pm t_{df,\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

Sample 1 Mean ($\overline{X_1}$)	7.0226
Sample 2 Mean ($\overline{X_2}$)	6.7806
Sample 1 Size (n_1)	647
Sample 2 Size (n_2)	53
Sample 1 Variance (S_1)	1.5803
Sample 2 Variance (S_2)	1.4309

Figure 15: Summary statistics of individuals between the two groups

Based on the table, degree of freedom $\frac{\left(\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{S_1^2}{n_1}\right)^2+\frac{1}{n_2-1}\left(\frac{S_2^2}{n_2}\right)^2} \approx 62$

$t_{critical}$ Value was calculated by using Excel, with a 5% significance level: T.INV.2T(0.05, 62)

$$t_{62,0.05/2} = \pm\ 1.9996$$

By substituting statistics into the confidence interval formula:

$$CI = 7.0226 - 6.7806 \pm 1.9996\sqrt{\frac{1.5803^2}{647} + \frac{1.4309^2}{53}}$$

$$CI = -\ 0.1011,\ 0.5851$$

$LCL = -\ 0.1011$

$UCL = 0.5851$

As a result, we have 95% confidence that the difference in relationship satisfaction between individuals in relationships for 7 years or fewer and those for more than 7 years, lies within the interval [-0.1011, 0.5851]

Accordingly, this reveals that the confidence interval includes zero, implying that implying the relationship satisfaction within the two samples is unlikely to be significantly impacted by the duration of a relationship. In essence terms, although there is a slight difference between samples, it is not large enough to confidently claim that satisfaction declines after seven years of relationship. Therefore, there is no strong statistical evidence to support the 7-Year Itch Hypothesis.

Hypothesis Test:

Null hypothesis(H_0)**:** There is no significant difference in relationship satisfaction between individuals in relationships of 7 years or fewer and those in relationships longer than 7 years.
Alternative hypothesis (H_1)**:** There is a significant difference in relationship satisfaction between individuals in relationships of 7 years or fewer and those in relationships longer than 7 years.

Given: $t_{statistic} = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\frac{s_1}{n_1} + \frac{s_2}{n_2}}} = 1.4104$ and $degree\ of\ freedom \approx 62$

The p-value was calculated by using Excel 2*T.DIST.RT(1.4104, 62)

$$p - value = 0.1635$$

Since $p - value = 0.1635 > 0.05$. There is insufficient evidence to reject the null hypothesis, indicating there is no significant relationship between individuals in relationships of 7 years or fewer and those in relationships longer than 7 years.

Conclusion:
By testing the 7-Year Itch Hypothesis using a test for unknown and unequal variances at the 5% significance level, the data was deduced to reveal insignificant differences in relationship satisfaction between those in relationships 7 years or less, and more than 7 years. We have used the findings of critical values, inclusive of zero, to further demonstrate that there is no strong evidence to show a decline in satisfaction based on relationship duration. In addition, the p-value was found to be 0.1635, which fails to reject the null hypothesis, thus creating insufficient evidence to support the 7-Year Itch Hypothesis.

C. Migrant vs Non-Migrant Comparison

This analysis addresses whether migrants and non-migrants factor influence relationship satisfaction with the utilisation of a two-sample separate-variance test, assuming unequal variances.

Null hypothesis (H_0**):** There is no difference in average relationship satisfaction between migrants and non-migrants.

$$H_0: \mu_1 = \mu_2$$

Alternative hypothesis (H_1**):** There is a difference in average relationship satisfaction between the migrants and non-migrants.

$$H_1: \mu_1 \neq \mu_2$$

μ_1: *population mean of relationship satisfaction among migrants*

μ_2: *population mean of relationship satisfaction among non-migrants*

Group	Sample size (n)	Sample mean ($\overline{x}$)	Sample standard deviation (s^2)
Migrants	179	7.0694	1.1715
Non-migrants	521	6.9819	1.2807

Figure 16: Sample statistics for migrants and non-migrants' relationship satisfaction

Separate-Variance t-test

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

By substituting statistics into the separate-variance t-test formula, we obtained the :test statistic.

$$t_{statistic} = 0.8416$$

P-Value Calculation:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{S_2^2}{n_2}\right)^2} \approx 335$$

The two-sided p-value was calculated using T.DIST.2T(0.8416, 335) on Excel we found that $p \approx 0.4006$. Thus, we fail to reject H_0, as p-value = $0.4006 > 0.05$.

T critical Value:

To assist in the decision of rejecting or retaining H_0, we conducted a two-sample hypothesis test for the difference in means, using a 5% significance level. Using the excel formula of T.INV.2T(0.05, 335), we found that null hypothesis is rejected if :

$$T_{CV} > 1.9671 \text{ or } T_{CV} < 1.9671$$

Since $t_{statistic} = 0.8416$, which lies within the acceptance region, there is insufficient evidence to reject the null hypothesis. Therefore, by using the p-value approach in alignment with the t-test, there is insufficient evidence to reject H_0 , allowing for the conclusion that there is no difference in relationship satisfaction between migrants and non-migrants.

Confidence Interval: To evaluate the effect of migrants and non-migrants on relationship satisfaction, a 95% confidence interval was constructed:

$$CI = (\overline{x_1} - \overline{x_2}) \pm t_{critical} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$CI = (7.0694 - 6.9819) \pm 1.9671 \times \sqrt{\frac{1.1715^2}{179} + \frac{1.2807^2}{521}}$$

$$CI = [-0.1170, 0.2921]$$

$LCL = -0.1170$

$UCL = 0.2921$

The confidence interval includes zero; it indicates that we can be 95% confident that there is no significant difference in relationship satisfaction between migrant and non-migrant relationship satisfaction.

D. Simple Regression: Urban-Rural Divide

This section investigates whether individuals living in urban locations report different levels of satisfaction compared to those in rural locations, by employing a simple linear regression model, in which the satisfaction score is the dependent variable, and residence location is the independent variable ("Urban"=1, "Rural"=0).

<u>Regression equation:</u>

$$Satisfaction = \beta_0 + \beta_1(Urban) + \epsilon$$

Where:

β_0: $the\ average\ satisfaction\ levels\ in\ rural\ areas$

β_1: $the\ average\ difference\ in\ satisfaction\ for\ urban\ residents$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.0019							
R Square	0.000003779							
Adjusted R Square	-0.0014							
Standard Error	1.2543							
Observations	700							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	0.0041	0.0041	0.0026	0.9591			
Residual	698	1098.1443	1.5733					
Total	699	1098.1485						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	7.0017	0.0685	102.1701	0.0000	6.8672	7.1363	6.8672	7.1363
Urban	0.0049	0.0949	0.0514	0.9591	-0.1815	0.1912	-0.1815	0.1912

Figure 17: Simple Regression Summary Statistics

According to the statistics from the table, the Regression Equation is given as:

$$Satisfaction = 7.0017 + 0.0049(Urban)$$

The positive coefficient of 'Urban' ($\beta_1 = 0.0049$) suggests that people in urban areas report 0.0049 points higher relationship satisfaction than those in rural areas.

Since $p - value = 0.9591$, it is greater than the significance level of 0.05, there is insufficient evidence to reject the null hypothesis. Consequently, there is no evidence of a statistically significant relationship satisfaction gap between urban and rural populations.

The R-squared value identifies how well the independent variable explains or can predict the data of the dependent variable. Given $R^2 = 0.000003779$, the ≈ 0, the model can explain 0.00% of the data variability in demonstrating the influence of relationship satisfaction and living locations. The statistics indicating whether a couple lives in an urban or rural area do not help predict their relationship satisfaction, suggesting there are a range of other factors affecting relationship satisfaction.

E. Multiple Regression Model

Model selection: To select the most appropriate model, we conducted a multiple regression model analysis, with satisfaction as a dependent variable, and nine 9 independent variables, comprised of gender, urban/rural residence, age, years in relationship, conflicts per week, time together per week, number of children, marriage status, and migrancy status.

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	4.503864339	0.288558054	15.60817407	2.93327E-47	3.937307149	5.07042153	3.937307149	5.07042153
Gender	-0.200256877	0.077585648	-2.581107241	0.010053687	-0.352589158	-0.047924596	-0.352589158	-0.047924596
Urban	0.030821668	0.077712091	0.396613547	0.691775091	-0.121758872	0.183402208	-0.121758872	0.183402208
Age	0.055506998	0.007770444	7.143349444	2.31938E-12	0.040250446	0.07076355	0.040250446	0.07076355
Years in Relationship	-0.029936781	0.020376377	-1.469190609	0.142236862	-0.069943922	0.01007036	-0.069943922	0.01007036
Conflict	-0.291975153	0.02757066	-10.59006768	2.19106E-24	-0.346107607	-0.237842699	-0.346107607	-0.237842699
Time together per week	0.102625827	0.007939374	12.92618685	2.20958E-34	0.087037597	0.118214056	0.087037597	0.118214056
Children	-0.017195579	0.083892418	-0.204971788	0.837654616	-0.181910622	0.147519465	-0.181910622	0.147519465
Married	0.310766544	0.079110219	3.928273076	9.41633E-05	0.155440907	0.46609218	0.155440907	0.46609218
Migrant	-0.050680933	0.089273777	-0.567702348	0.570421764	-0.225961781	0.124599915	-0.225961781	0.124599915

Figure 18: Multiple Regression Analysis of all 9 variables

Based on the table above, the p-values for urban (0.6918), years in relationship (0.1422), children (0.8377), and migrancy status (0.5704) are all larger than the significance level of 0.05. Thus, we excluded them from further analysis, and continued the conduction of the multiple regression analysis with gender, age, conflicts per week, time together per week, and marriage status as independent variables.

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	4.379777254	0.272627329	16.0650705	1.29994E-49	3.844503999	4.91505051	3.844504	4.91505051
Gender	-0.202583676	0.077407598	-2.617103243	0.009061084	-0.354564834	-0.050602518	-0.3545648	-0.0506025
Age	0.055478059	0.007751903	7.156701747	2.10763E-12	0.040258064	0.070698053	0.04025806	0.07069805
Conflict	-0.290236246	0.027410482	-10.58851317	2.17452E-24	-0.34405366	-0.236418832	-0.3440537	-0.2364188
Time together per week	0.102566459	0.007918027	12.9535374	1.58703E-34	0.087020299	0.118112619	0.0870203	0.11811262
Married	0.305597554	0.078842005	3.87607538	0.000116225	0.150800099	0.460395009	0.1508001	0.46039501

Figure 19: Multiple Regression Analysis of 6 variables

According to the table, it is evident that all the p-values of independent variables are smaller than 0.05, suggesting that they are all significant, indicating that this is a suitable multiple

regression analysis for relationship satisfaction. The formula of the regression model is formulated as follows:

$$\widehat{Y} = b_0 + b_1X_{1i} + b_2X_{2i} + b_3X_{3i} + b_4X_{4i} + b_5X_{5i}$$

$$\widehat{Y} = 4.3798 - 0.2026X_{Gender} + 0.0555X_{Age} - 0.2902X_{Conflict} + 0.1026X_{Time} + 0.3056X_{Married}$$

<u>Interpret the model coefficients:</u>

- **Intercept (4.3798)**: indicates that an individual who is female, has an age of zero, no conflict, spends zero hours with their partner per week, and is unmarried is expected to have a satisfaction score of 4.3798 points, assuming all other variables are held constant. While age zero is not practical in reality,, this value serves as a reference point for further deductions.

- **Gender (-0.2026)**: shows that males are expected to have 0.2026 points lower relationship satisfaction levels than females, holding all other variables constant.

- **Age (0.0555):** indicates that with each additional year of age, relationship satisfaction increases by 0.0555 points, holding all other variables constant.

- **Conflict (-0.2902):** shows that with each additional conflict per week, a relationship satisfaction decreases by 0.2902 points, holding all other variables constant.

- **Time (0.1026):** indicates that for each additional hour of time spent together per week, relationship satisfaction increases by 0.1026 points, holding all other variables constant.

- **Marriage (0.3056):** indicates that being married increases relationship satisfaction by 0.3056 points compared to those who are not married, assuming all other variables are constant.

<u>The overall fit of the model:</u>

The F-test of overall significance indicates whether a linear regression model provides a better fit to the data than a model that contains no independent variables.

Null Hypothesis: $H_0 = \beta_1 = \beta_2 = ... = \beta_6 = 0$

(There is no linear relationship between the dependent and the independent variables.)

Alternative Hypothesis: $H_1 =$ at least one $\beta_1 \neq 0$

(There is a linear relationship between the dependent and at least one of the independent variables.)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.58250676
R Square	0.3393
Adjusted R Square	0.3346
Standard Error	1.022464673
Observations	700

ANOVA

	df	SS	MS	F	Significance F
Regression	5	372.6172969	74.5235	71.28470931	3.36524E-60
Residual	694	725.5312018	1.0454		
Total	699	1098.148499			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	4.3798	0.272627329	16.0650705	1.29994E-49	3.844503999	4.91505051	3.844504	4.91505051
Gender	-0.2026	0.077407598	-2.617103243	0.009061084	-0.354564834	-0.050602518	-0.35456483	-0.05060252
Age	0.0555	0.007751903	7.156701747	2.10763E-12	0.040258064	0.070698053	0.04025806	0.07069805
Conflict	-0.2902	0.027410482	-10.58851317	2.17452E-24	-0.34405366	-0.236418832	-0.34405366	-0.23641883
Time together per week	0.1026	0.007918027	12.9535374	1.58703E-34	0.087020299	0.118112619	0.0870203	0.11811262
Married	0.3056	0.078842005	3.87607538	0.000116225	0.150800099	0.460395009	0.1508001	0.46039501

Figure 20: Multiple Regression Summary Statistics

Based on the Multiple Regression Summary Statistics, the F-statistic is calculated as:

$$F_{STAT} = \frac{MSR}{MSE} = \frac{74.5235}{1.0454} = 71.2847$$

Critical Value:

Significance level of $\alpha = 0.05$

$$df_1 = k = 9, df_2 = n - k - 1 = 700 - 9 - 1 = 690$$

By using Excel, we get $F_{0.05} = F.INV(0.95,\ 9,\ 690) = 1.8934$

Since $F_{STAT}(71.2847)$ is greater than the critical value $F_{0.05}\ (1.8934)$, we reject H_0..

P-Value (ANOVA Table). Since the p-value of 3.36524E-60 is lower than 0.05, we can reject H_0.

Conclusion: There is sufficient evidence that at least one independent variable affects relationship satisfaction, I.e. at least one $\beta_1 \neq 0$.

The model fit:

- R-squared (0.3393): This means that 33.93% of the variation in relationship satisfaction can be explained by this model. Explained by the independent values: gender, age, conflicts per week, time together per week, and marriage marital status.

- The adjusted R-squared (0.3346): This 33.46% value is adjusted for the number of predictors in the model. This slightly lower value of R-squared is negligible, with the similarity representing that the model is not overfitted, and each of the chosen independent variables has a meaningful contribution to relationship satisfaction.

Residual Analysis:

Figure 21: Residual Plot of Gender, Age, Conflict, Time Spent, and Married

Linearity: Since the residual plots are generally scattered around zero, supporting the assumption of linearity and represents the relationship between independent and dependent variables.

Independence: Since this data set was not gathered over a period of time, the autocorrelation is not likely to occur.

Normality: Since the sample contains 700 individuals, it is large enough to implement the Central Limit Theorem, which tends to be normal distribution even if the residuals are not perfectly distributed through the plot. Therefore, the normality assumption is met.

Equal Variance: It is evident that the points are scattered around zero, indicating no systematic or wide or narrow model. It is evenly spread as the x-axis increases, suggesting the equal variances and homoscedasticity assumption.

Scatter Plot Analysis:

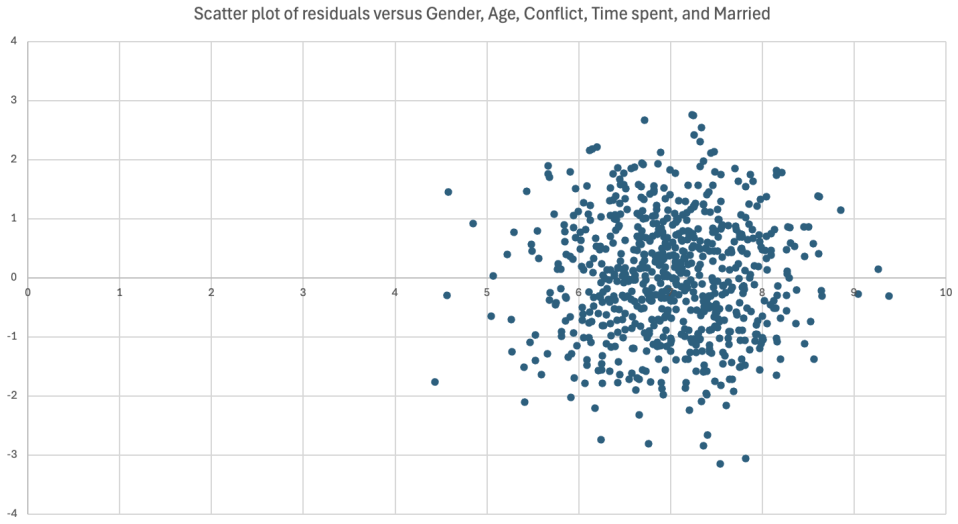

Figure 22: Scatter plot of residuals versus Gender, Age, Conflict, Time spent, and Marriage

As can be observed, the scatter plot is centralised and spread around the horizontal axis, not exhibiting any strong pattern or trends. The absence of clustering further suggests that the assumptions of linearity, homoscedasticity, and lack of autocorrelation are met.

F. Revisiting the Gender, Age, Conflict, Time together per week, and Married

In part B, we came to the conclusion that there is insignificant evidence to support the 7-Year Itch hypothesis, with no clear indication between satisfaction levels and relationship duration, whether they are less than seven years or more than seven years long, supported by a 95% CI range from -0.1011 to 0.5851, which includes zero. In the multiple regression model, the p-value=0.1422>0.05, has been excluded from the model, illustrating that by holding other factors constant, the relationship duration does not affect the individual's satisfaction.

Additionally, in part C, we conducted the two-sample t-test, finding t-statistic=0.8416, p-value=0.4006 and the 95% CI range from -0.1170 to 0.2921, including zero, which indicates that the difference between satisfaction among migrants and non-migrants is not significant. In part E, the $p - value = 0.5704 > 0.05$, which has hence been excluded. Migrant status does not impact relationship satisfaction, as there is no significant difference in satisfaction between migrants and non-migrants, holding every other variable constant.

In part D, by calculating $p - value = 0.9591$ and $R^2 \approx 0$, the effect of rural-urban on relationship satisfaction was found to be insignificant. As $p - value = 0.6918$ is larger than 0.05, it has been excluded from the multiple regression model. The urban-rural residency factors do not influence relationship satisfaction. While the simple regression in part D only accounts for the urban-rural factor, the multiple regression analysis performed in part E takes into account a range of other variables, such as gender, age, conflicts, time together per week, and marriage.

The R-squared value in part E (33.93%) is significantly higher than the R-squared found in part D (0.00%), suggesting that the model is a lot more accurate when using the multiple regression model. Therefore, it is best to use the multiple regression model to understand the key factors that affect relationships among Australians.

Summary

The report shows no statistically significant difference in satisfaction between migrants and non-migrants, suggesting that based solely on migration background does not determine relationship satisfaction. It is also evident that the relationship length does not affect the relationship satisfaction. We also assessed the urban–rural residents factor, but found no meaningful difference in satisfaction levels between the two groups when other variables were held constant. By conducting a multiple regression analysis, factors such as gender, age, conflict, time together per week, and marriage have a far stronger impact on relationship satisfaction than the individual's living location and their migration background.

Accordingly, these findings suggest that effective relationship wellbeing initiatives, Centre for Australian Relationship Excellence should prioritise reducing conflict, encouraging quality time between partners, supporting marriage and long-term commitment, addressing in specific gender relationship needs, given the lower satisfaction among men, and considering age factor in relationship dynamics rather than focusing solely on background demographics. These initiatives will help CARE foster healthier and more fulfilling relationships across Australia.

Contribution Statement

The team met weekly to review progress and address any challenges. All members actively participated, analysed the data, and contributed to the final presentation.

Group Members	Tasks
Tulga Altai	Supporting part E, proofread.
Kelsey Jia	Part A.
Nina Lu	Part D, proofread, making slides.
Hien Pham	Part B, E, F, summary, supporting in Part A, and C, making slides.
Aneya Sarkar	Part C, proofread, making slides.

AI Statement

We acknowledge the use of Grammarly to refine the academic language of our work. We submitted this report with a prompt to improve the tone and accuracy of language, including grammatical structures, punctuation, and vocabulary. The output was then modified further to better represent our own tone and style of writing.

We acknowledge the use of OpenAI to analyse data of our work. We submitted this report with a prompt to improve the tone and accuracy of language, including grammatical structures, punctuation, and vocabulary. The output was then modified further to better represent our own tone and style of writing.